

TrES-3b

R-2809

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120829 · created 2026-09-05T01:08:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456168.7035 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.051
Transit depth (Rp/Rs)^2	1.82 +/- 1.37 [%]
Orbital Inclination (inc)	81.93 +/- 2.63
Airmass coefficient 1 (a1)	1.022 +/- 0.02
Airmass coefficient 2 (a2)	-0.027 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.94 %
Best Comparison Star	#3 - [461, 475]
Optimal Method	PSF photometry
Transit Duration (day)	0.045 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.051 vs 0.16309 (deviation 0.55 σ)
Duration fit vs published	0.045 d vs 0.0567 d (deviation 0.43 σ)
Mid-time O-C	1.6 min
Depth SNR	2.6
Residual scatter	1.94 %

Light curve

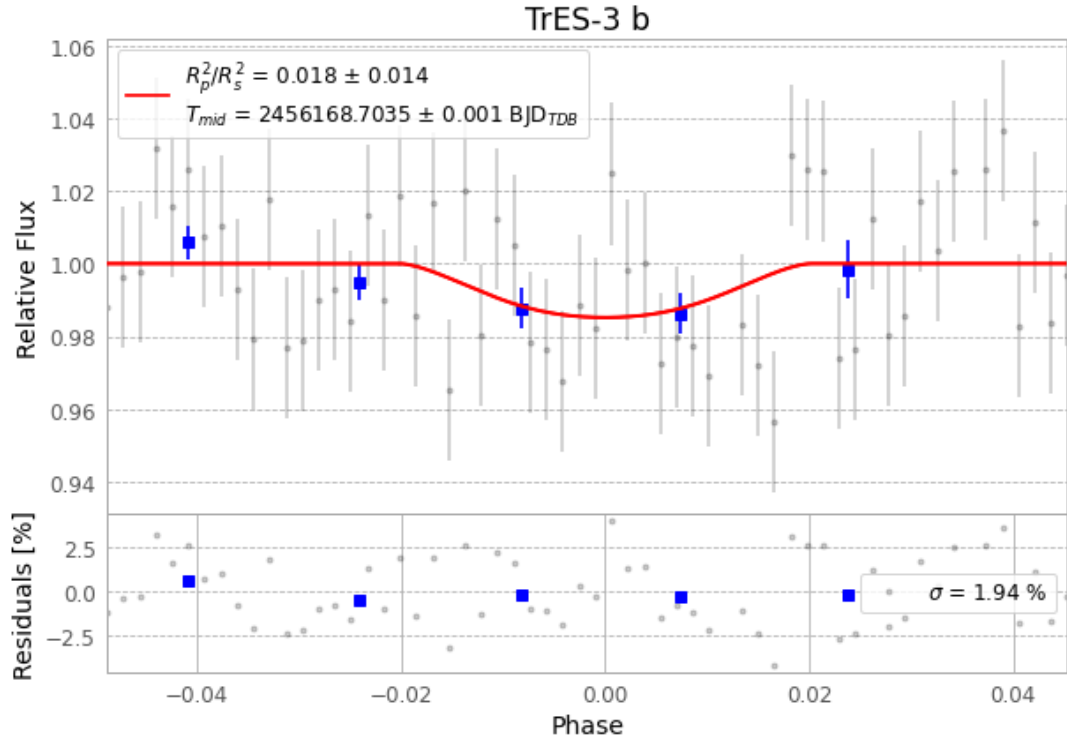


Figure 1 — FinalLightCurve_TrES-3 b_2012-08-28.png (SHA-256 a200077fe1ffdab9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [219, 265], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 22e931e88a3bb76d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

60 FITS frames (39 MB), TRES-3120829031813.FITS → TRES-3120829061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120829.log (923852f20393...), FinalLightCurve_TrES-3 b_2012-08-28.png (a200077fe1ff...), FinalLightCurve_TrES-3 b_2012-08-28.csv (1ffb40536f35...)

Compute resources

Wall time 357.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 01:08Z.